



Today's agenda

↳ Remove K from BST

↳ Serialization

↳ Deserialization

↳ Binary tree Cameras → HW



AlgoPrep



Q) Given K , Remove the Node from BST.

→ (i) removal of leaf node: remove the node.

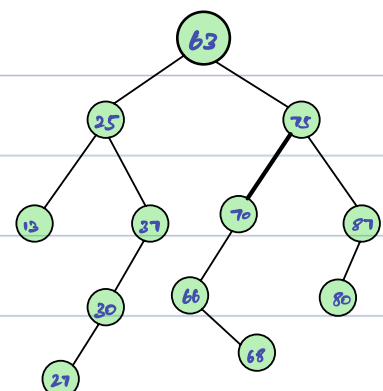
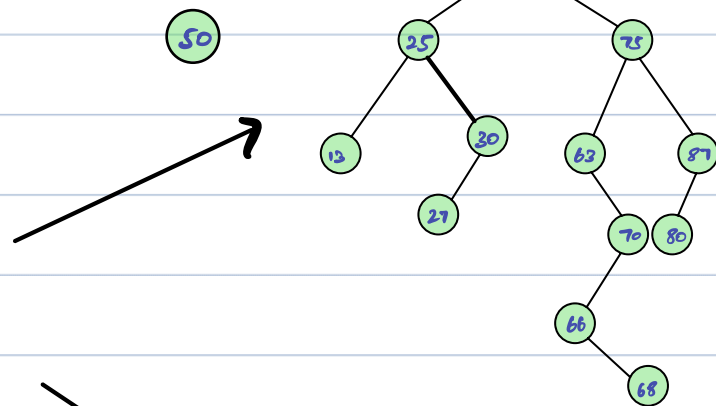
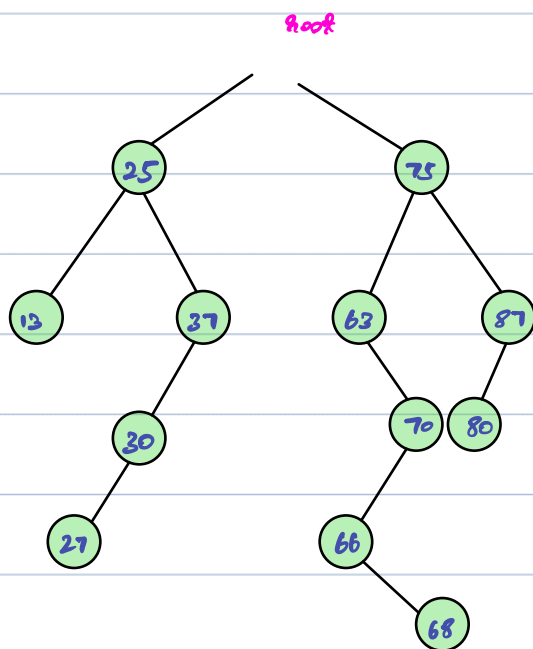
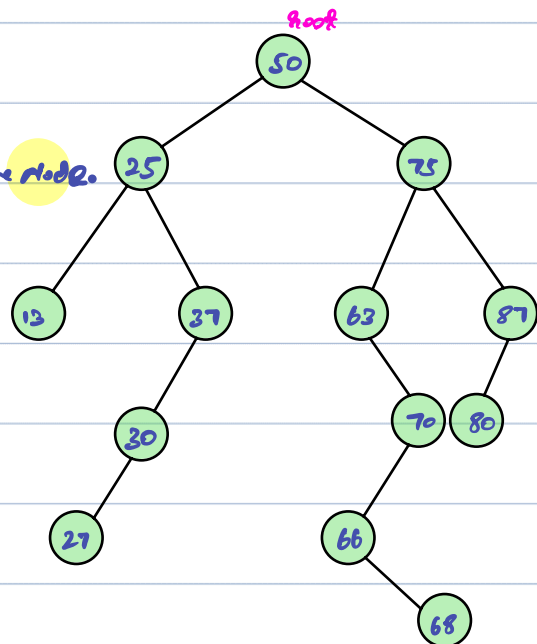
→ (ii) Node with left child:

↳ Remove and make the connection of Parent with the child.

→ (iii) Node with right child:

↳ Remove and make the connection of Parent with the child.

(iv) Node with both the child.





// Pseudo code

```
Node DeleteinBST (Node root, int k) {  
    if (root == null) { return null; }  
  
    if (root.val > k) {  
        root.left = DeleteinBST (root.left, k);  
    }  
    else if (root.val < k) {  
        root.right = DeleteinBST (root.right, k);  
    }  
    else {  
        if (root.left == null || root.right == null) {  
            return null;  
        }  
        else if (root.left != null && root.right == null) {  
            return root.left;  
        }  
        else if (root.left == null && root.right != null) {  
            return root.right;  
        }  
        else {  
            int max = func (root.left);  
            root.val = max;  
            root.left = DeleteinBST (root.left, max);  
        }  
    }  
}
```

T.C: $O(H)$

S.C: $O(H)$

return root;



3

3

return root;

3



AlgoPrep

Node DeleteInBST (Node root, int k){

if (root == null) { return null; }

if (root.val > k) {

root.left = DeleteInBST (root.left, k);

}

else if (root.val < k) {

root.right = DeleteInBST (root.right, k);

}

else {

if (root.left == null || root.right == null) {

return null;

}

else if (root.left != null || root.right != null) {

return root.left;

}

else if (root.left == null || root.right != null) {

return root.right;

}

else {

int max = func (root.left);

root.val = max;

root.left = DeleteInBST (root.left, max);

}

return root;

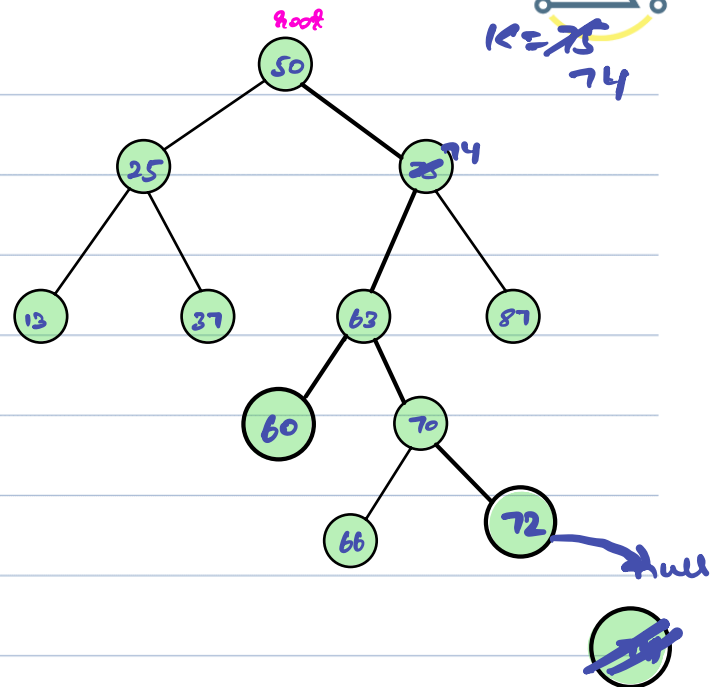
}

return root;

}



k = 75
74

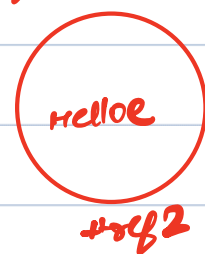


StringBuilder

String S1 = "Hello" → #ref1

String S2 = "Hello" → #ref1

S1 = S1 + "e" → #ref2



→ StringBuilder sb = new StringBuilder(S1);

↳ sb = sb.append("e");

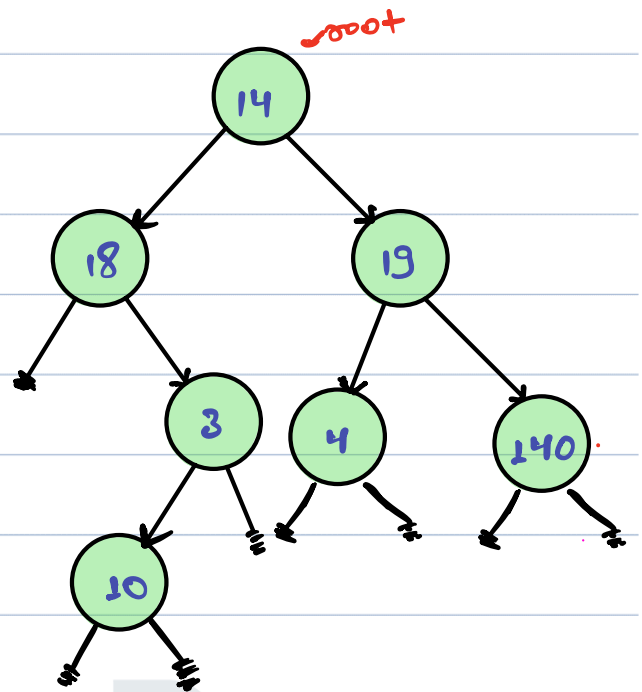
→ sb.toString();



Q) Serialize and Deserialize a Binary tree

↳ Can you convert a tree in linear form.

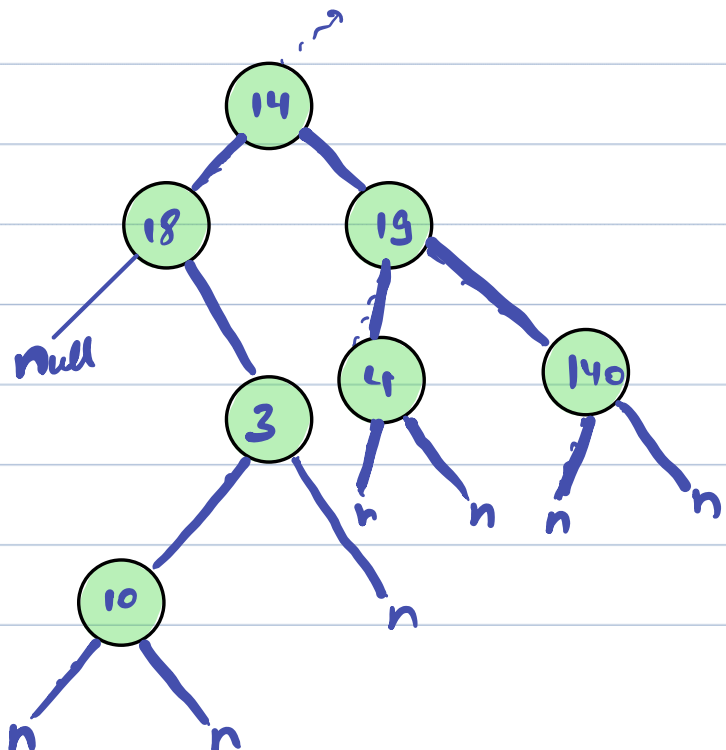
→ Sb = " "



Serialize: Do Preorder, add val with space and instead of null add # with space.

st: 14 _ 18 _ # _ 3 _ 10 _ # _ # _ # _ 19 _ 4 _ # _ # _ 140 _ # _ # _

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14
ch[i]: 14 18 # 3 10 # # # 19 4 # # 140 # #





// Pseudo Code

```
String serialize (Node root) {  
    StringBuilder sb = new StringBuilder();  
    Helper(root, sb);  
    return sb.toString();  
}
```

```
void Helper (Node root, StringBuilder sb) {  
    if (root == null) {  
        sb.append("# ");  
        return;  
    }  
    sb.append(root.val + " ");  
    Helper(root.left, sb);  
    Helper(root.right, sb);  
}
```

T.C: $O(N)$

S.C: $O(H)$

```
int i;  
Node Deserialize (String data) {  
    String[] ch = data.split(" ");  
    i = 0;  
    Node root = Helper(ch);  
    return root;  
}
```



```
Node Helper2 (String[] ch){  
    if (ch[i].equals("#")){  
        i++;  
        return null;  
    }  
}
```

```
Node root = new Node (Integer.parseInt(ch[i]));  
i++;
```

```
root.left = Helper2 (ch);
```

```
root.right = Helper2 (ch);
```

```
return root;
```

```
}
```